Passive RF Localization of Non-Cooperative Drones

Mathematics, Algorithms and System Architecture

dronelocate — technical reference

August 8, 2026

Abstract

This document describes the theory and implementation of a passive time-difference-of-arrival (TDOA) localization system for non-cooperative drones. Ten sensor nodes observe the drone's own radio emissions, stream compressed IQ to a central supernode over a publish/subscribe bus, and the supernode cross-correlates the captures, solves a three-dimensional position by weighted nonlinear least squares, and tracks it with a Kalman filter. The treatment covers coordinate frames and dilution of precision, the signal and clock models, detection (energy and cyclostationary), time-delay estimation (band-limited interpolation, the cross-ambiguity function, and generalised cross-correlation weighting), the position solver (grid-seeded Gauss–Newton with robust loss, an altitude prior, and generalised least squares over all node pairs), and the tracking filter including tightly coupled updates from under-determined bursts. A final section collects the numerical-conditioning failures encountered during development, several of which cost more accuracy than any algorithmic choice discussed here.

Contents

1 Problem statement

A drone that transmits — video downlink, control uplink, telemetry — can be located from its own emissions, with no cooperation and no transmission from the sensor network. This is the passive, or *silent*, case: the sensors radiate nothing and are therefore neither detectable nor jammable by the target.

Let the emitter be at unknown position $\mathbf{x} \in \mathbb{R}^3$ and let sensor node i occupy known position $\mathbf{n}_i \in \mathbb{R}^3$. A signal leaving the emitter at unknown time t_0 arrives at node i at

$$t_i \;=\; t_0 \;+\; \frac{\|\mathbf{x} - \mathbf{n}_i\|}{c} \;+\; \varepsilon_i, \tag{1}$$

where c is the speed of light and ε_i is node i's clock error relative to a common timebase. The emission time t_0 is unknown and uninteresting, so it is eliminated by differencing pairs of arrivals:

$$\tau_{ij} \;=\; t_i - t_j \;=\; \frac{\|\mathbf{x} - \mathbf{n}_i\| - \|\mathbf{x} - \mathbf{n}_j\|}{c} \;+\; (\varepsilon_i - \varepsilon_j). \tag{2}$$

Each measured τ_{ij} constrains $\mathbf{x}$ to one sheet of a hyperboloid of revolution with foci $\mathbf{n}_i, \mathbf{n}_j$. Three independent constraints fix a point in space; more than three over-determine it and permit a least-squares solution with a covariance.

Two facts drive the entire design. First, $c \approx 0.3\,\mathrm{m/ns}$, so a timing error of one nanosecond is thirty centimetres of range error — and the residual clock term $\varepsilon_i - \varepsilon_j$ in (2) enters with exactly the same weight as geometry. Timing discipline, not signal processing, sets the achievable accuracy. Second, τ_{ij} must be *measured*, which requires the same emission to be recognisable in two different receivers' samples; that measurement is a cross-correlation, and everything in Section 6 exists to make it unbiased.

2 System architecture

2.1 Topology and data flow

Figure 1 shows the deployment. Nodes are thin: they capture, detect, buffer, and announce. The supernode holds all the geometry and makes all the decisions.

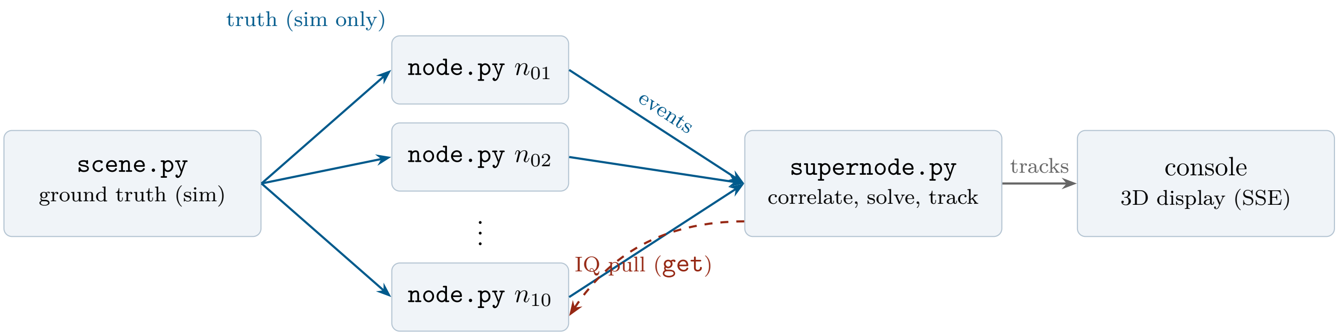

Figure 1: Star topology. The supernode listens; nodes connect as clients. Solid arrows are small, frequent, never-dropped messages; the dashed arrow is the bulk IQ transfer, which is *pulled* on demand.

2.2 Pull, do not push

A node hears a burst, computes a detection event of roughly one kilobyte, and publishes that. The IQ itself — 47 kB per 10 ms snippet at 2.4 Msps in `ci8` — stays in the node's ring buffer. The supernode groups the events belonging to one burst, chooses a geometrically favourable subset of the nodes that heard it, and issues a query only to those.

Ten nodes typically hear a burst and six or seven are queried. Halving the transfers costs nothing in accuracy because the discarded nodes were geometrically redundant — the selection criterion in Section 3.2 makes that statement precise.

2.3 Quality of service

The bus carries traffic with incompatible requirements, so each key expression is assigned a priority and a congestion policy:

Key expression	Priority	Congestion	Purpose
`dc/{site}/{node}/evt/detect`	interactive high	block	$\sim$1 kB events
`dc/{site}/{node}/iq`	background	block	IQ queryable (pull)
`dc/{site}/{node}/health`	data low	drop	telemetry
`dc/{site}/cmd/{node}`	interactive high	block	retasking
`dc/{site}/track/live`	data high	block	fused tracks
`dc/{site}/sim/truth`	—	—	**simulation only**

The priority split is what stops a 47 kB bulk transfer from delaying a one-kilobyte detection event. Health telemetry *drops* rather than blocking, because a stale health message is worthless and must never apply backpressure to the pipeline.

Design invariant. Nothing on the solve path reads the ground-truth channel. Emulated nodes subscribe to it to render *their own* IQ — each applying its own geometric delay, clock error, path loss and noise — then discard it and publish only detections. The supernode does subscribe, but exclusively to score the answer in metres for the display. Delete that key expression and the system still localizes, merely unscored. This is what makes the simulation an honest test rather than a demonstration.

The same separation governs the display. The console can drape OpenStreetMap raster tiles and extruded building footprints over the survey plane, and can re-centre that basemap on any of several cities. All of it is decoration: no measurement is taken against it, the buildings are drawn at a default height wherever the source has no surveyed one, and the map's coordinate conversion is a flat tangent-plane approximation rather than the ellipsoidal transform of Section 3. Re-centring the basemap moves the streets under the constellation, not the constellation — sensor positions are metres in a local frame and do not depend on which city is drawn beneath them.

3 Coordinate frames and geometry

3.1 WGS84 to local tangent plane

All estimation runs in a local East–North–Up (ENU) Cartesian frame in metres. Geodetic coordinates appear only at the configuration boundary and at the display boundary. Least squares performed in degrees introduces a direction-dependent bias, because a degree of longitude and a degree of latitude are different distances, and that anisotropy would be absorbed silently into the position estimate.

Geodetic to Earth-Centred Earth-Fixed, with semi-major axis a, flattening f, and $e^2 = f(2-f)$:

$$N(\varphi) = \frac{a}{\sqrt{1 - e^2 \sin^2 \varphi}}, \tag{3}$$

$$\mathbf{p}_{\mathrm{ECEF}} = \begin{bmatrix} (N+h)\cos\varphi\cos\lambda \\ (N+h)\cos\varphi\sin\lambda \\ \big(N(1-e^2)+h\big)\sin\varphi \end{bmatrix}. \tag{4}$$

With a site origin $(\varphi_0, \lambda_0, h_0)$ the rotation into ENU is

$$\mathbf{R} = \begin{bmatrix} -\sin\lambda_0 & \cos\lambda_0 & 0 \\ -\sin\varphi_0\cos\lambda_0 & -\sin\varphi_0\sin\lambda_0 & \cos\varphi_0 \\ \cos\varphi_0\cos\lambda_0 & \cos\varphi_0\sin\lambda_0 & \sin\varphi_0 \end{bmatrix}, \qquad \mathbf{p}_{\mathrm{ENU}} = \mathbf{R}\,(\mathbf{p}_{\mathrm{ECEF}} - \mathbf{p}_{0,\mathrm{ECEF}}). \tag{5}$$

The inverse uses Bowring's iteration, which converges to sub-millimetre at these altitudes in a handful of steps.

3.2 Dilution of precision

Linearising (2) about a candidate position gives the measurement Jacobian. With $\hat{\mathbf{u}}_i$ the unit vector from the target toward node i, the row for the pair (i, ref) is $\hat{\mathbf{u}}_{\mathrm{ref}} - \hat{\mathbf{u}}_i$. Collecting the rows into $\mathbf{H}$, the geometric covariance factor is $(\mathbf{H}^\top \mathbf{H})^{-1}$, whence

$$\mathrm{HDOP} = \sqrt{C_{11} + C_{22}}, \qquad \mathrm{VDOP} = \sqrt{C_{33}}, \qquad \mathrm{PDOP} = \sqrt{\operatorname{tr} \mathbf{C}}, \qquad \mathbf{C} = (\mathbf{H}^\top \mathbf{H})^{-1}. \tag{6}$$

Position error is approximately DOP $\times\, c\,\sigma_t$. For the reference ten-node layout, HDOP ≈ 0.59 while VDOP ≈ 16.4. That ratio is not a defect of the solver: with eight of ten nodes near ground level, every measurement row is nearly horizontal and the vertical component of $\mathbf{x}$ is weakly observable. Two nodes are deliberately elevated (45 m and 70 m) to break the coplanarity; setting them back to ground level roughly triples the vertical error. *Vertical accuracy is bought with mast height, not with software.*

3.3 Node selection

Given the set $\mathcal{H}$ of nodes that heard a burst, the supernode chooses a subset $\mathcal{S} \subseteq \mathcal{H}$ of bounded size that minimises PDOP. The search is greedy: seed with the strongest node by SNR, then repeatedly add whichever candidate most reduces PDOP evaluated at a provisional target position. That position is an RSSI-weighted centroid,

$$\mathbf{x}_{\mathrm{seed}} = \frac{\sum_{i \in \mathcal{H}} w_i \mathbf{n}_i}{\sum_{i \in \mathcal{H}} w_i}, \qquad w_i = 10^{\,\mathrm{RSSI}_i/20}, \tag{7}$$

with the vertical component replaced by the configured altitude prior. The seed only has to be good enough to *rank* geometries, and RSSI is already present in the event, so it costs nothing.

How much geometry is enough. Selection only pays if adding sensors actually buys dilution, and for this constellation it stops paying early. Evaluated at the orbit centre, over every subset containing an elevated node:

Sensors	HDOP	VDOP	PDOP
all 10	0.62	10.83	10.85
best 7	0.83	11.05	11.08
best 6	0.99	11.11	11.16
best 5	3.21	12.41	12.82

Seven of ten is indistinguishable from ten, six is close behind, and five falls off a cliff — HDOP nearly quadruples, because at five the survivors no longer surround the target and the horizontal problem becomes one-sided. The vertical barely moves throughout, which is the same point as Section 3.2 from the other direction: VDOP here is set by the sensors' *elevation spread*, not by how many of them there are, so adding ground-level sensors cannot fix it.

Two practical consequences. First, the cap on $|\mathcal{S}|$ is not a compromise: pulling IQ from more than about seven well-chosen nodes buys nothing measurable, which is what makes the pull-not-push architecture of Section 2.2 free rather than lossy. Second, and less obviously, *configuring* more emulated nodes than the cap is pure waste — the surplus render a full signal chain, run detection and publish events that are never queried. On a constrained machine that is around a third of the simulation's cost for no geometric return.

4 Signal and channel model

The simulator exists to test the estimator, so it must reproduce every effect that can bias the estimator — and be honest about the ones it does not model.

4.1 Emitted waveform

Each node synthesises the emitted burst independently from the burst identifier, so all nodes agree on the waveform without ever exchanging it. Two waveform families are available.

Band-limited Gaussian noise. A filtered complex Gaussian burst has a near-ideal thumbtack autocorrelation, so correlation quality tracks SNR rather than waveform structure. This isolates solver behaviour, and is the right choice when the question is about geometry or timing.

CP-OFDM. The realistic case. A drone video downlink is OFDM, so the synthetic emitter builds N_{sym} symbols with $N_u = 128$ useful samples and a cyclic prefix of $N_{cp} = 16$:

$$s_m[n] = \frac{1}{N_u} \sum_{k \in \mathcal{K}} X_{m,k}\, e^{j2\pi kn/N_u}, \qquad n = -N_{cp}, \ldots, N_u - 1, \tag{8}$$

with $X_{m,k}$ QPSK on the active carriers $\mathcal{K}$, pilots at fixed positions every eighth carrier, and a DC null. At 2.4 Msps this is 18.75 kHz carrier spacing, adjacent to the LTE family that commercial drone links resemble.

The distinction matters more than it appears. *A flat spectrum makes whole classes of technique inexpressible*: spectral weighting (Section 6.5) degenerates to a constant, and cyclostationary detection (Section 5.2) has no cyclic structure to find. An algorithm validated against white noise is validated against nothing.

Pilots must be scrambled, and the reason is a delay-estimation hazard rather than a communications one. Repeating an identical pilot vector on every symbol makes the waveform periodic at the symbol rate, which places autocorrelation sidelobes at multiples of $N_u + N_{cp}$ at some 17% of the main peak. Whether that matters depends entirely on the sample rate: at 2.4 Msps one symbol is 60 μs and the sidelobes fall outside a $\pm 30\,\mu$s search window, while at 20 Msps a symbol is 7.2 μs and they fall well inside it — where a correlator will occasionally lock onto one and return a confident lag wrong by an exact multiple of the symbol period. The pilots therefore carry a per-symbol BPSK scrambling sequence, as real OFDM does. Per-subcarrier power is unchanged, so the spectrum keeps its pilot structure for weighting, and the cyclic prefix is untouched, so the detector of Section 5.2 is unaffected; measured sidelobes fall roughly twentyfold. Note the general form of the lesson: *a narrowband test cannot see this defect at all*, because the artefact only enters the search window once the capture is wide.

4.2 Frequency hopping

A real datalink does not sit still. It hops across the band on a schedule the receiver is not told, so the emitter transmits at $f_c + \delta_b$ for burst b, and in a node's baseband the burst appears at δ_b rather than at DC. Energy shifted beyond the digitized band never reaches the ADC at all — the analog front end removes it — so a hop outside the window does not arrive attenuated, it does not arrive.

That distinction is what makes hopping more dangerous than any effect considered so far. Every other impairment in this document degrades a measurement; this one deletes it, and a missed detection cannot be recovered downstream by weighting, gating or robust loss. Measured against a 17.5 MHz hop span, a 2.4 Msps node detects 1 burst in 16, with a median post-hop SNR of −106 dB. Section 5.3 is the receiver-side answer.

In the simulator the hop is derived from the burst identifier, so every node renders the same hop without exchanging anything — the same coordination-free construction as the waveform itself — while the receiver is told nothing and must find the energy for itself.

4.3 Multipath

Free-space propagation gives each node exactly one arrival. A real site adds reflections, so node i observes

$$y_i[n] = \sum_{\ell=0}^{L} a_{i\ell}\, s(n - \Delta_i - f_s\, \tau_{i\ell}) + \nu_i[n], \qquad \tau_{i0} = 0, \tag{9}$$

with excess delays $\tau_{i\ell} > 0$ drawn from an exponential power-delay profile and total echo power a Rician K below the direct path. The reflectors are buildings and terrain, so the profile is a property of the node's location: it is drawn once and held, not redrawn per burst. That choice is the whole point. A per-burst redraw would average away over a run and appear as noise; a static profile biases every measurement from that node in the same direction, forever.

The consequences are developed in Section 6.6, but the signature is worth stating with the model: multipath produces a large *bias* while leaving *precision* untouched. The measurement is exactly as repeatable as it was before, and simply wrong.

4.4 Propagation and receiver

Node i's baseband capture over a window beginning at t_{cap} is

$$y_i[n] = A_i\, g[n]\, s(n - \Delta_i)\, e^{\,j(2\pi f_{d,i} n/f_s + \phi_i)} + \nu_i[n] + \iota_i[n], \tag{10}$$

with the fractional sample delay

$$\Delta_i = f_s \Big[(t_0 - t_{\text{cap}}) + \frac{r_i}{c} + \varepsilon_i \Big], \qquad r_i = \|\mathbf{x} - \mathbf{n}_i\|. \tag{11}$$

Here A_i is free-space amplitude ($\propto 1/r_i$), $g[n]$ a raised-cosine gate placing the burst inside the middle 40% of the window, ν_i complex Gaussian noise, and ι_i an optional node-local interferer. Fractional delay is applied exactly, as a phase ramp in the frequency domain: $\mathcal{F}\{s(n-\Delta)\}(f) = S(f)\, e^{-j2\pi f \Delta}$.

The gate is not cosmetic. A window that is entirely signal has no noise floor to measure against, and an SNR estimator formed from mean and median over such a window returns approximately 0 dB — a detector that silently detects nothing.

4.5 Clock discipline

The clock error $\varepsilon_i(t)$ is modelled in three regimes, selectable at run time.

GPSDO (disciplined). A disciplined oscillator is a control loop, so its error is *bounded* noise about UTC rather than a ramp. It is modelled as an Ornstein–Uhlenbeck process with correlation time τ and stationary deviation σ_{pps}, discretised exactly:

$$e_k = \alpha\, e_{k-1} + \sqrt{1-\alpha^2}\, \sigma_{\text{pps}}\, \eta_k, \qquad \alpha = e^{-\Delta t/\tau}, \quad \eta_k \sim \mathcal{N}(0,1), \tag{12}$$

and the total error is $\varepsilon_i = b_i + e_k$. The static bias b_i survives discipline: GPS corrects phase and rate against UTC, but antenna cable delay and receiver group delay are calibration constants no amount of locking removes. A bias *common* to all nodes cancels in (2); the per-node part does not, and recovering it is exactly what reference-emitter calibration would do.

Holdover. Lock is lost; the oscillator free-runs from wherever its error stood at that moment, at its own drift rate: $\varepsilon_i(t) = e_{\text{hold}} + \delta_i(t - t_{\text{hold}})$.

Free. Never disciplined: an unbounded ramp $\varepsilon_i(t) = b_i + \delta_i(t - t_{\text{ref}})$. At $\delta_i = 0.02$ ppm this reaches 24 μs in twenty minutes — which is how the correlation-window failure of Section 6.1 was found.

4.6 Carrier, Doppler and local-oscillator offset

A real receiver downconverts from f_c, and two effects put a frequency offset on the resulting baseband. Emitter motion gives a range rate $\dot{r}_i = (\mathbf{x} - \mathbf{n}_i) \cdot \mathbf{v}/r_i$ and hence $\dot{\tau}_i = \dot{r}_i/c$; the node's own oscillator contributes a fractional rate error ρ_i. Together,

$$f_{d,i} \;=\; -f_c\,(\dot{\tau}_i + \rho_i) \;=\; -\frac{f_c}{c}\,\frac{(\mathbf{x} - \mathbf{n}_i) \cdot \mathbf{v}}{r_i} \;-\; f_c\,\rho_i. \tag{13}$$

The magnitude decides the architecture. A quadrotor at 31 m/s seen at 2.437 GHz produces a *differential* offset between a node pair of 250–400 Hz. Over a $T = 10$ ms window, $\Delta f\,T \approx 2.5$–4. Section 6.4 shows that a plain cross-correlation is attenuated by $\operatorname{sinc}(\Delta f\,T)$, so the peak is not merely degraded but past its first null.

Why the carrier matters and the sample clock does not. With $f_c/f_s \approx 10^3$, the same fractional error is a thousand times larger in carrier phase than in sample timing. Over 10 ms, a 0.02 ppm clock skews sampling by 0.05 samples — negligible — while the carrier turns several full cycles. This is why the model includes Doppler and LO offset but deliberately omits sample-clock skew.

4.7 Wire formats

Uplink bandwidth, not computation, dominates the latency budget, so IQ is quantised before transmission:

Format	B/sample	10 ms snippet	Cost
`cf32`	8	188 kB	reference
`ci8`	2	47 kB	negligible; native to most hardware
`ci2`	0.5	12 kB	≈ 0.55 dB correlation SNR

Two-bit quantisation is the VLBI trick. Correlation is an averaging operation, so coarse amplitude information is nearly free to discard; what matters is timing. On a constrained uplink, `ci2` is worth more than a GPU.

5 Detection

5.1 Energy detection

Split the window into blocks, form the mean power of each, and compare the strongest block against the *median* block:

$$\widehat{\mathrm{SNR}}_{\mathrm{dB}} = 10\log_{10}\frac{\max_b P_b - \operatorname{median}_b P_b}{\operatorname{median}_b P_b}. \tag{14}$$

The median is the noise-floor estimator rather than the mean because the mean is dragged upward by the very burst being measured.

5.2 Cyclostationary (cyclic-prefix) detection

An OFDM symbol repeats its own tail: the cyclic prefix is a copy of the last N_{cp} samples of the useful part. Therefore $y[n]$ and $y[n+N_u]$ are identical wherever n lies inside a prefix. The statistic

$$\rho(N_u) \;=\; \max_{\text{windows } W} \frac{\left|\sum_{n\in W} y[n]\, y^*[n+N_u]\right|}{\sum_{n\in W} |y[n]|^2} \tag{15}$$

is compared against a threshold $\kappa/\sqrt{|W|}$ over a small set of candidate useful lengths. Two properties make this valuable:

1. **It is immune to frequency offset.** Any carrier or Doppler term $e^{j2\pi f_d n/f_s}$ contributes $e^{j2\pi f_d n/f_s} e^{-j2\pi f_d (n+N_u)/f_s} = e^{-j2\pi f_d N_u/f_s}$, a constant phase that leaves $|\rho|$ untouched. The detector therefore does not care that the target is moving — unlike an energy detector following a fading channel, and unlike correlation, which cares a great deal (Section 6.4).
2. **It exploits structure, not power.** Measured on the synthetic OFDM emitter, ρ crosses threshold at -2 dB SNR where the energy gate requires $+8$ dB: a margin of 10 dB, which is a substantial increase in detection range.

The statistic is evaluated over sliding windows rather than the whole capture because a global mean dilutes it by the burst's duty cycle. Once N_u is known, the strongest spectral line of the product sequence $y[n]y^*[n+N_u]$ gives the symbol period $N_u + N_{cp}$, hence the prefix length — a classifier, since prefix geometry distinguishes signal families.

5.3 Acquisition in frequency: channelization

Detection presumes the signal is inside the digitized band. Against the hopping emitter of Section 4.2 that presumption fails fifteen times in sixteen, so the node must digitize the whole hop span and locate the burst before any of the preceding machinery applies.

Channelize by FFT into N_c sub-bands and score each one against its *own* temporal noise floor rather than against its neighbours:

$$\hat{c} = \arg\max_{c} \frac{\max_b P_c[b]}{\operatorname{median}_b P_c[b]}, \tag{16}$$

with $P_c[b]$ the power in channel c during block b. Scoring against the channel's own floor rather than absolute power is the same reasoning as the block-power detector: a strong narrowband interferer parked in one channel would win an absolute-power comparison, and loses this one because it is always on.

The chosen channel is then shifted to DC, low-pass filtered and decimated, after which every stage already described — cross-ambiguity, spectral weighting, sub-sample interpolation, the solver — runs unchanged on a narrowband stream. Decimation also divides the uplink burden, since only the channel that mattered is shipped. Measured, the same 17.5 MHz span that defeats a narrowband node is recovered in full: 16 of 16 bursts, with every node independently agreeing on the channel.

The channel grid is a coordination mechanism, not a convenience. Nodes snap the downconversion to a shared grid rather than each estimating the true hop centre. A per-node estimate would be more accurate in isolation and would break the system: two nodes would then downconvert to slightly different frequencies, and that *differential* offset — up to half a channel step, 1.25 MHz in the measured configuration — is three orders of magnitude beyond what the Doppler search of Section 6.4 can absorb. Snapping to a grid lets independently operating nodes agree with no message exchange, exactly as the burst-seeded waveform and the scheduled-capture grid do. Nodes that nevertheless disagree did not hear the same signal, and must be dropped rather than correlated across.

Channelization and bandwidth pull in opposite directions. Decimating to isolate a hop is correct when the target is narrowband and costs nothing, because the retained band still contains the whole signal. Applied to a *wideband* target it discards the signal instead of isolating it: measured on an 18 MHz emitter decimated to 2.5 MHz, horizontal error degraded from 0.26 m to 418 m. Bandwidth is what buys timing precision (Section 6.3) and multipath resolution (Section 6.6); channelization buys coverage of a hopper. The receiver must refuse any configuration that decimates below the signal bandwidth, since the failure is otherwise silent.

6 Time-delay estimation

This is where the accuracy is won or lost. Everything downstream treats τ_{ij} as data.

6.1 Cross-correlation and the search window

For captures a and b the cross-correlation is computed by FFT:

$$R_{ab}[k] \;=\; \mathcal{F}^{-1}\{\mathcal{F}\{a\}\cdot\overline{\mathcal{F}\{b\}}\}[k]. \tag{17}$$

Only lags inside a physically plausible window are searched. Restricting the search suppresses sidelobe peaks that would otherwise win at low SNR, so the window should be as narrow as correctness permits.

The subtlety is where to *centre* it. The peak sits at geometric TDOA *plus* the clock offset difference, and only the first term is bounded by the array baseline. Clock error accumulates from session start, so a window centred on zero and sized by geometry eventually excludes the true peak; the correlator then locks onto noise and returns a confident wrong answer. The window is therefore *translated* to the expected offset $c_0 = \lfloor f_s\,\Delta\varepsilon\rceil$ rather than widened:

$$k^\star = \underset{|k-c_0|\le K}{\arg\max}\,|R_{ab}[k]|, \qquad K = \left\lceil f_s\Big(\frac{B_{\max}}{c} + 5\,\mu\text{s}\Big)\right\rceil, \tag{18}$$

with $B_{\max}$ the longest baseline. Widening would also work but would sacrifice the sidelobe rejection the narrow window buys.

A quality figure accompanies each measurement: the peak divided by the RMS of the searched window with the peak neighbourhood excised,

$$q \;=\; \frac{|R_{ab}[k^\star]|}{\sqrt{\mathrm{mean}_{|k-k^\star|>3}\,|R_{ab}[k]|^2}}. \tag{19}$$

6.2 Sub-sample interpolation

The correlation peak almost never falls on a sample. The estimator's resolution therefore depends entirely on how the peak location is refined, and here the obvious choice is wrong.

Fitting a parabola through three magnitude samples assumes the peak is locally quadratic. It is not: the correlation is band-limited, so its value between samples is a *sinc interpolation* of its neighbours. With $B/f_s \approx 0.83$ the peak is oversampled barely 1.2× and its main lobe spans about one sample — a three-point fit has almost nothing to grip on. Measured against known fractional delays, parabolic interpolation on magnitude carries a systematic S-curve bias reaching 0.12 samples (≈ 50 ns, ≈ 15 m of range) that reverses sign across the sample. Being systematic, it does not average away and no covariance rescaling can absorb it.

The implementation reconstructs the band-limited complex correlation directly:

$$\hat{R}(k^\star+\theta) \;=\; \sum_{m=-M}^{M} R_{ab}[k^\star+m]\;\mathrm{sinc}(\theta-m)\,w[m], \qquad \theta\in[-\tfrac12,\tfrac12], \tag{20}$$

with w a Blackman taper suppressing the ringing of the truncated kernel, evaluated on a dense grid and refined by one parabolic step *on that grid*. Measured accuracy is 0.0017 samples (0.7 ns), a 46× improvement over the parabola. Jacobsen's complex three-point estimator was also tried and is worse here (0.091 samples) for the same reason: at this oversampling ratio no three-point fit has enough information.

6.3 Measurement uncertainty

Each correlation is assigned a timing standard deviation combining the theoretical bound with the interpolator floor:

$$\sigma_t \;=\; \max\left(\frac{1}{2\pi B\sqrt{\widehat{\mathrm{SNR}}}},\; \frac{1}{256 f_s}\right), \qquad \widehat{\mathrm{SNR}} = q^2 - 1. \tag{21}$$

The floor matters: when it was set by the parabolic interpolator at 1/16 of a sample, it returned the same 26 ns for every quality above about 4, hiding the difference between a clean peak and a marginal one from the solver.

6.4 The cross-ambiguity function

Section 4.6 established that two nodes observe the same emission at different frequencies. Write the differential offset as Δf. The correlation of the two captures over a window of length T then carries the factor

$$\left|\frac{1}{T}\int_0^T e^{j2\pi\Delta f\, t}\, dt\right| = |\mathrm{sinc}(\Delta f\, T)|, \tag{22}$$

so the peak collapses as $\Delta f\, T$ approaches unity, and inverts beyond it. Before it collapses it *biases*: a partially decorrelated peak still clears the quality gate while sitting at the wrong lag, which is the dangerous regime.

The remedy is to search both dimensions — the cross-ambiguity function

$$\chi(\tau,\nu) \;=\; \int_0^T a(t)\,\overline{b(t-\tau)}\, e^{-j2\pi\nu t}\, dt, \qquad (\hat\tau,\hat\nu) = \arg\max_{\tau,\nu} |\chi(\tau,\nu)|. \tag{23}$$

Evaluating this on a dense grid is expensive, so it is factored. Split the capture into S segments of length L, correlate each segment pair (batched FFTs), and observe that the peak's phase advances linearly from segment to segment at exactly the differential frequency. One FFT *across* the segment index therefore yields the whole Doppler axis:

$$Z[p,k] = \sum_{m=0}^{S-1} R_{ab}^{(m)}[k]\; e^{-j2\pi pm/S}, \qquad \hat\nu = \frac{p^\star}{S\,T_{\mathrm{seg}}}. \tag{24}$$

The frequency estimate is refined parabolically, applied as a derotation of b, and the *lag* is then measured by the full-window correlator of the previous subsections — so none of the sub-sample accuracy is sacrificed to the coarse segmentation. A zero Doppler estimate reproduces the plain correlator exactly, which is asserted in the test suite.

Segment length trades resolution against ambiguity: S segments span $\pm f_s/(2L)$ unambiguously, so S is chosen from the configured maximum target speed and the operating frequency.

6.5 Generalised cross-correlation weighting

Generalised cross-correlation applies a real weight $W(f)$ to the cross-spectrum before inversion:

$$R_{ab}^{\mathrm{GCC}}(\tau) = \int W(f)\, S_{ab}(f)\, e^{j2\pi f\tau}\, df. \tag{25}$$

The maximum-likelihood (Hannan–Thomson) choice is

$$W_{\mathrm{HT}}(f) = \frac{1}{|S_{ab}(f)|} \cdot \frac{|\gamma(f)|^2}{1 - |\gamma(f)|^2}, \tag{26}$$

with γ the magnitude-squared coherence. Since single-snapshot coherence is badly biased, it is built here from per-channel SNR spectra: with each channel's noise floor estimated as a low quantile of its Welch PSD, $s_a = S_a/\hat{n}_a - 1$ and

$$|\gamma|^2 = \frac{s_a s_b}{(1+s_a)(1+s_b)}. \tag{27}$$

The textbook weight was measured and rejected. The factor $1/(1 - |\gamma|^2)$ in (26) carries the ML pedigree and also the failure mode: at any SNR worth localizing, in-band coherence saturates, the factor rails against whatever regularisation bounds it, and the weight degenerates into phase-transform (PHAT) whitening. Measured on an interfered node pair: 4.0 ns RMS against plain correlation's 7.2 ns — worse than useless, since it also costs peak quality. Dropping the factor and keeping

$$W(f) \;=\; \frac{|\gamma(f)|^2}{\sqrt{S_a(f)S_b(f)}}$$

— coherence to null the bins only one node trusts, SCOT division to crush interference by its own power — gives 2.5 ns on the same data.

The regime where this pays is *node-local* interference: a neighbouring emitter visible to one node has no cross-channel coherence, so $|\gamma|^2 \approx 0$ in its bins and the weight deletes it. Interference common to all nodes retains coherence and survives; separating that case requires true cross-spectral coherence estimated over segments, which in turn requires per-pair Doppler derotation first.

A counter-intuitive consequence deserves emphasis: weighted correlations report *lower* quality by (19) while being more accurate, because the weight suppresses the very sidelobe floor that the quality ratio divides by. Detection thresholds must never be retuned against weighted quality figures.

6.6 Multipath, and why bandwidth is the only defence

Under the channel of Section 4.3 the correlation is no longer a single peak but a superposition, and the estimator returns the composite maximum rather than the direct-path arrival. Two facts determine what can be done about it.

The error is bias, not noise. Measured across node pairs at 2 MHz with $K = 8$ dB and $\tau_{\mathrm{rms}} = 200$ ns, the median |bias| rises from 1.1 to 12.3 ns (worst 60 ns) while the run-to-run standard deviation is *unchanged* at 0.4 ns. Across the fleet this is $0.21 \rightarrow 4.57$ m horizontally and $2.1 \rightarrow 114$ m vertically — the vertical hit some 25× harder because VDOP ≈ 16 amplifies each node's static offset.

Neither existing defence removes it. The reduced chi-square rises from 0.19 to 94.9, so the system does register that something is wrong, and (33) inflates the covariance roughly tenfold — but not enough: containment collapses from 100% to 0%. Honest degradation, still under-reported. Robust loss does nothing at all ($4.57 \rightarrow 4.50$ m), and its failure is instructive rather than disappointing: Huber weights are scaled by the *median* residual spread, so they identify a measurement out of line with its peers. Multipath biases every node, so there is no outlier to find. A defence built on relative disagreement cannot see a common-mode error.

Bandwidth resolves it. A reflection is separable only when its excess delay exceeds one resolution cell, c/B. At $B = 2\,\text{MHz}$ that cell is 150 m (500 ns) and echoes at 80–400 ns are inside it: they do not form a second peak, they distort the only one. At $B = 18\,\text{MHz}$ the cell is 17 m (56 ns), the echoes separate, and the same scenario yields 0.26 m horizontal and 35 m vertical — a seventeenfold improvement from bandwidth alone. This is the same lever as (21), and it is the strongest argument for pointing a wideband receiver at a drone's video downlink rather than its narrow control link.

Leading-edge detection is the textbook answer and it does not work. Since echoes arrive strictly later than line-of-sight, taking the first credible peak rather than the largest is the standard remedy. It was implemented and measured, and removed. Three results, in ascending order of cost. Unbounded, it is catastrophic: the search window spans the array's geometric extent, tens of microseconds, while an echo is at most a microsecond or two late, so the search accepted an unrelated structural peak 21 μs early and placed the fix 158 km away — any first-arrival search must be bounded to a plausible excess delay before the maximum. Bounded, it is a no-op wherever the direct path is strong, because the maximum already *is* the direct path and the residual bias comes from echoes too close to resolve. And in the shadowed case where it should win, it traded horizontal error $38 \to 15$ m for vertical $38 \to 520$ m, because inconsistent per-node peak choices are precisely what a large VDOP magnifies. Multipath is a bandwidth problem and a siting problem, not a peak-picking problem.

7 Position solution

7.1 Measurement model and cost

For a reference node ref and measurement set $\mathcal{M}$, the residual of measurement i at candidate position $\mathbf{x}$ is

$$r_i(\mathbf{x}) = \frac{\|\mathbf{x} - \mathbf{n}_i\| - \|\mathbf{x} - \mathbf{n}_{\text{ref}}\|}{c} - \tau_i, \tag{28}$$

and the weighted least-squares cost, including the altitude prior, is

$$J(\mathbf{x}) = \sum_{i \in \mathcal{M}} w_i\, r_i^2(\mathbf{x}) \;+\; \frac{(x_3 - \bar{h})^2}{\sigma_h^2}, \qquad w_i = \sigma_{t,i}^{-2}. \tag{29}$$

The altitude prior is a soft constraint, not a fudge. Section 3.2 showed the vertical is weakly observable; without a prior, vertical error leaks into the horizontal solution through the off-diagonal covariance. The same prior is applied again in the tracker.

7.2 Grid seed

The TDOA cost surface is multimodal — hyperboloid sheets intersect in more than one place — so a Gauss–Newton iteration started at the array centroid can converge to a ghost intersection and report it confidently. J is therefore evaluated first on a coarse three-dimensional grid ($21 \times 21 \times 9$ over the site and a plausible altitude band) and the minimum taken as the starting point. This costs one vectorised pass and removes the entire class of ghost-lock failures.

7.3 Damped Gauss–Newton

With $d_i = \|\mathbf{x} - \mathbf{n}_i\|$ and $\hat{\mathbf{e}}_i = (\mathbf{x} - \mathbf{n}_i)/d_i$, the Jacobian rows are

$$\mathbf{J}_i = \frac{\hat{\mathbf{e}}_i - \hat{\mathbf{e}}_{\text{ref}}}{c}, \tag{30}$$

and each iteration solves the Levenberg-damped normal equations

$$(\mathbf{J}^\top \mathbf{W} \mathbf{J} + \lambda\, \mathrm{diag}(\mathbf{J}^\top \mathbf{W} \mathbf{J}))\, \delta\mathbf{x} = -\,\mathbf{J}^\top \mathbf{W} \mathbf{r}, \tag{31}$$

accepting the step only if the cost decreases (halving λ) and otherwise increasing λ fourfold. Prior rows are appended to both sides.

7.4 Robust loss

A node with a mis-surveyed position correlates beautifully and lies consistently; a gate cannot see it. Huber IRLS bounds its influence:

$$u_i = |r_i|\sqrt{w_i}, \qquad s = 1.4826 \ \mathrm{median}_i \, u_i, \qquad \psi_i = \min\left(1, \ \frac{k}{u_i/s}\right), \quad k = 1.345, \tag{32}$$

with weights recomputed from the current residuals each iteration. The scaling by the *median* residual spread rather than by the assumed σ is load-bearing: when the assumed sigmas are optimistic, every normalised residual is large and a fixed threshold down-weights all measurements equally — a rescale, not outlier rejection. Dividing by the observed spread asks the right question, namely whether a measurement is out of line *with its peers*. Measured with one node corrupted by an $8\,\mu$s lag error: plain least squares 330.6 m, robust 8.9 m.

7.5 Covariance and its calibration

The naive covariance $(\mathbf{J}^\top \mathbf{W} \mathbf{J})^{-1}$ reports how confident the *assumed* sigmas were, not how well the fit agrees with the data. Operators trust the displayed ellipsoid, so it must degrade when the solution degrades. The measurement information is therefore inflated by the reduced chi-square before the prior is added:

$$\chi^2_{\mathrm{red}} = \frac{\sum_i \psi_i w_i r_i^2}{n-3}, \qquad \mathbf{A} = \frac{\mathbf{J}^\top \mathbf{W} \mathbf{J}}{\max(\chi^2_{\mathrm{red}}, 1)} + \frac{1}{\sigma_h^2} \operatorname{diag}(0,0,1), \qquad \mathbf{C} = \mathbf{A}^{-1}. \tag{33}$$

The order of operations is essential. Scaling the finished covariance instead would inflate the prior's contribution too — but the prior is not a measurement, and its uncertainty does not grow because the TDOAs disagree with one another. On the vertical axis, where the prior dominates, that mistake turned a 60 m prior into a 160 m reported sigma and made σ_v *anti*-correlate with the actual error: a high chi-square made the solver lean harder on the prior, landing it near the true altitude while claiming to be less sure. The ellipsoid grew tallest exactly when the fix was best.

7.6 All pairs and generalised least squares

Correlating every node against a single reference gives $n-1$ measurements whose errors share that reference's timing noise. Weighting them as independent over-counts the reference's information. Correlating *all* pairs gives $\binom{n}{2}$ measurements whose covariance is structured and known:

$$\mathbf{R} = \mathbf{A} \ \operatorname{diag}(\mathbf{s}) \, \mathbf{A}^\top + \sigma^2_{\mathrm{interp}} \mathbf{I}, \tag{34}$$

where $\mathbf{A} \in \{0, \pm 1\}^{m \times n}$ is the pair incidence matrix (row k has $+1$ in column i, -1 in column j) and s_i is node i's timing variance. The per-node variances are not observed directly, but with many pairs the system $v_k \approx s_i + s_j$ is over-determined and a non-negative least-squares split recovers them from the measured pair sigmas — no new model, only accounting.

The solution then minimises the properly whitened cost $\mathbf{r}^\top \mathbf{R}^{-1} \mathbf{r}$, implemented by solving $\mathbf{L}\mathbf{u} = \mathbf{r}$ with $\mathbf{L}$ the Cholesky factor of $\mathbf{R}$ and running the same damped Gauss–Newton on $\mathbf{u}$. Robust weights apply to the *whitened* residuals for the same reason.

Measured over deterministic bursts, mean horizontal error falls from 4.33 m (star reference) to 3.54 m (all pairs), with the gain concentrated on the worst bursts, which is where correlated reference noise does its damage. The feared $3.5\times$ correlation cost was eliminated by caching one forward FFT per node and correlating from the cached spectra; Doppler derotation becomes an integer-bin circular shift of a cached spectrum, since $\mathcal{F}\{b[n]e^{j2\pi\kappa n/N}\} = (\mathcal{F}\{b\})[\,\cdot - \kappa\,]$.

7.7 Outlier defences

Two mechanisms operate on different failure modes and are both enabled by default.

Quality gate. Discard a correlation whose peak fails to stand out, $q < q_{\min}$. This is *not* redundant with σ_t: by (21) a correlation that locked onto pure noise — wrong by microseconds, therefore by kilometres — receives a sigma within a factor of two of a clean peak's. The sigma model assumes the peak found is the right peak merely blurred; it has no term for *this is a different peak entirely.* Only a gate expresses that.

Robust loss. As above, for outliers a gate cannot see.

A warning about tuning: the gate must not be tightened to improve χ^2. Because χ^2 is computed over the survivors, an over-tight gate flatters its own metric while starving the solve. Measured, raising $q_{\min}$ from 8 to 45 cut χ^2_{red} from 17.7 to 5.9 and made the position error *worse*, from 26.1 to 33.1 m, while discarding a third of all fixes.

8 Direction finding from coherent channels

A receiver with two or more coherent channels — sharing a local oscillator and an ADC clock — measures a bearing from phase alone, with no inter-node timing whatsoever. This is a fundamentally different observable from TDOA and it fails in different circumstances, which is exactly why it is worth having.

8.1 Two-element interferometry

For elements separated by d and a plane wave arriving at angle θ from the array normal, the inter-element phase difference is

$$\Delta\phi = \frac{2\pi d}{\lambda}\sin\theta \quad\Longrightarrow\quad \theta = \arcsin\!\left(\frac{\lambda\,\Delta\phi}{2\pi d}\right), \tag{35}$$

with $\Delta\phi$ estimated from the complex inner product of the two channels,

$$\Delta\hat{\phi} = \arg\!\left(\sum_n \overline{a[n]}\,b[n]\right) - \phi_{\text{cal}}, \qquad \text{coherence} = \frac{\left|\sum_n \overline{a[n]}\,b[n]\right|}{\|a\|\,\|b\|}. \tag{36}$$

The argument order is not cosmetic: swapping a and b conjugates the result and mirrors every bearing about the array normal — a sign error that produces entirely plausible output while placing targets on the wrong side.

8.2 Ambiguity is a property of the array

Equation (35) inverts uniquely only when $d \le \lambda/2$ (61 mm at 2.44 GHz). For wider spacing the phase wraps, and the critical point is that it wraps *while* $|\sin\theta|$ *remains within unity* — so a range check on the answer cannot detect the ambiguity. All angles consistent with the measurement,

$$\theta_k = \arcsin\!\left(\frac{(\Delta\hat{\phi} + 2\pi k)\lambda}{2\pi d}\right), \qquad k \in \mathbb{Z},\ \ |\cdot| \le 1, \tag{37}$$

are therefore enumerated and reported as a candidate set rather than silently collapsed to one value. The true angle is always in the set; resolving it requires a third element, a wider aperture at a different spacing, or cross-node consistency. Two elements additionally cannot separate front from back, a cone ambiguity that survives any spacing.

Wider spacing buys angular precision and costs uniqueness. Measured on synthetic signals, a coherent pair achieves 0.02° standard deviation at 20 dB SNR and 0.25° at 0 dB.

8.3 Why it complements TDOA

Bearing error grows linearly with range while timing error does not, so DF alone degrades with distance in a way TDOA does not. In exchange it needs no inter-node synchronisation at all, which means it survives GPS jamming and spoofing — a realistic threat in precisely the scenarios this system addresses. That, and not accuracy, is its value.

> **Bearings will not repair the vertical axis.** It is tempting to assume DF helps where the geometry is weakest, but a planar array measures azimuth in its own plane and carries no elevation information: it implicitly assumes the source lies on the horizon. Published guidance for a five-element circular array is explicit that elevation estimation with such an array "is probably not a good idea", with tracking documented to fail above roughly 45° elevation — because the wavefront then cuts a foreshortened and distorted aperture. A drone overhead sits precisely in that regime. Fusing bearings to shore up VDOP would therefore feed the weakest axis a measurement that degrades exactly where it is needed, and would require either an elevation-dependent σ or a non-planar array.

Fusing bearings as additional rows of (29) remains worthwhile *for GPS-denied operation*, and the measurement already travels on the bus: a dual-channel node ships its bearing, candidate set, ambiguity flag and coherence in the detection event. Nothing consumes them yet. When it does, the better construction is to carry the whole angular power spectrum as a likelihood term rather than a chosen peak plus a candidate list — the solver's timing rows then resolve the grating-lobe ambiguity of Section 8 implicitly, instead of the node having to commit to an answer it cannot justify.

9 Tracking

9.1 Constant-velocity Kalman filter

The state is $\mathbf{z} = [\mathbf{p}^\top, \mathbf{v}^\top]^\top \in \mathbb{R}^6$ in ENU with the usual constant-velocity transition

$$\mathbf{F}(\Delta t) = \begin{bmatrix} \mathbf{I}_3 & \Delta t\, \mathbf{I}_3 \\ \mathbf{0} & \mathbf{I}_3 \end{bmatrix}, \qquad \mathbf{Q} = \mathrm{diag}(\tfrac{\Delta t^4}{4}\boldsymbol{a},\ \Delta t^2 \boldsymbol{a}), \quad \boldsymbol{a} = (a_h^2, a_h^2, a_v^2). \tag{38}$$

Process noise is deliberately anisotropic ($a_h = 6$, $a_v = 1.5\,\mathrm{m/s^2}$): a quadrotor manoeuvres hard laterally and climbs slowly, and an isotropic $\mathbf{Q}$ lets the weakly observed altitude random-walk until the filtered track is worse than the raw fixes feeding it. A fix updates the filter linearly, since the measurement *is* a position; the altitude prior is reapplied as a scalar pseudo-measurement.

A linear filter suffices because the grid seed inside the solver has already resolved the multimodality; by the time a fix reaches the tracker the posterior is unimodal.

9.2 Tightly coupled updates

A burst yielding fewer than three usable correlations cannot produce a fix — but it is not information-free. Each surviving pair is still a hyperboloid constraint, and an existing track supplies the prior the burst lacks. Such bursts are applied directly as an extended Kalman update with

$$h_k(\mathbf{z}) = \frac{\|\mathbf{p} - \mathbf{n}_{a_k}\| - \|\mathbf{p} - \mathbf{n}_{b_k}\|}{c}, \qquad \mathbf{H}_k = \left[\ \frac{\hat{\mathbf{e}}_{a_k} - \hat{\mathbf{e}}_{b_k}}{c}\ \middle|\ \mathbf{0}_{1\times 3}\ \right], \tag{39}$$

linearised at the predicted position and gated on the normalised innovation squared,

$$\epsilon = \mathbf{y}^\top \mathbf{S}^{-1} \mathbf{y}\ \le\ \chi^2_{0.99}(k), \qquad \mathbf{S} = \mathbf{H}\mathbf{P}\mathbf{H}^\top + \mathbf{R}, \tag{40}$$

so a single bad pair causes the track to coast rather than corrupt. No track is ever born from under-determined data. Measured: two rows pull a 31 m coasted track to 5.7 m, while an 8 μs outlier row is rejected.

9.3 Data association

Association is nearest-track with a 2.5 km gate, which is adequate for a single target. Multiple simultaneous drones require joint probabilistic data association; this is a known limit rather than an oversight.

10 Processing flow

Figure 2 traces one burst from radio to display.

11 Numerical conditioning

Several of the largest accuracy gains in this system's history came not from better algorithms but from removing arithmetic that destroyed information before the algorithms ever saw it. They are collected here because they generalise.

11.1 Absolute timestamps in a difference

The delay (11) was originally formed by building an absolute arrival instant $t_0 + r_i/c + \varepsilon_i$ and subtracting the capture start afterwards. At $t \approx 1.785 \times 10^9$ s the spacing of `float64` is

$$\mathrm{ulp}(t) = 2^{\lfloor \log_2 t \rfloor - 52} \approx 2.38 \times 10^{-7}\,\mathrm{s} = 238\,\mathrm{ns}. \tag{41}$$

Time of flight and clock error — the quantities the entire system exists to measure, both far smaller than 238 ns — were therefore rounded onto a 238 ns grid before the subtraction could recover them. The cost was roughly 70 ns RMS per node, 100 ns between a pair, some 30 m of range, discarded for nothing.

The fix is to form the difference of the two large, nearby epochs *first*, exactly as in (11): $(t_0 - t_{\mathrm{cap}})$ is computed without loss because the operands are close, and every small quantity is added afterwards.

A synthetic test that starts its clock at zero cannot see this. The unit test used $t_0 = 0$, and so had no epoch magnitude to lose precision against; it reported 0.8 ns residual while the live pipeline sat at 58 ns. Any test of a timing system must run at realistic absolute times.

11.2 Drift anchored to the wrong epoch

Accumulating oscillator drift from the Unix epoch rather than from session start gives $\delta \times 1.78 \times 10^9\,\mathrm{s} \approx 35\,\mathrm{s}$ of apparent clock error. That is not a subtle inaccuracy: it turned a 10 ms capture into an 85-million-sample allocation and wedged the node's callback thread. Drift must accumulate from a session reference.

11.3 Evaluating a stochastic model twice

Under the disciplined-clock model the error is a stateful process. Calling the model a second time to populate event metadata therefore reported an error the signal never had; the supernode subtracted the wrong number and the difference was pure uncorrectable error of about 21 ns. A stochastic model must be evaluated once per event and the value reused. The bug was harmless under the older deterministic model, which is precisely why it appeared the moment realism was added.

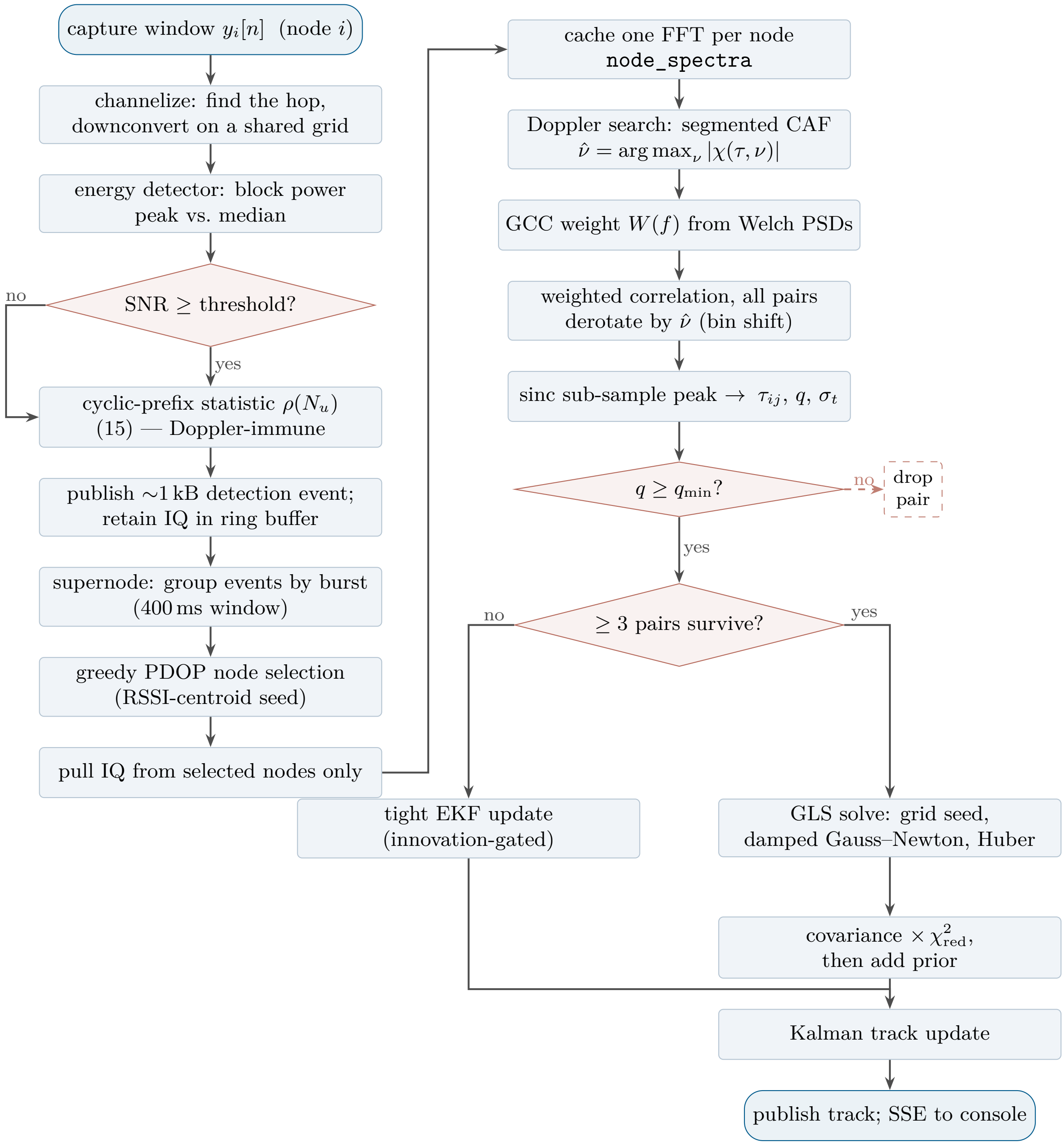

Figure 2: Per-burst processing. The left column runs on each sensor node; the right column runs on the supernode. Note the two recovery paths that a naive pipeline would omit: a burst failing the energy gate is still admitted when the cyclostationary statistic fires (Section 5.2), and a burst with too few surviving pairs to solve still updates an existing track through the tightly coupled filter rather than being discarded.

11.4 Scoring against the wrong reference

The error metric compared each fix against the most recently received ground-truth message rather than the truth of the burst that fix solved. Once solve latency exceeded the truth interval, the metric measured *latency*, not accuracy: at 31 m/s and two bursts per second a 150 ms solve reads as 15.8 m of horizontal error that no solver change can remove. The diagnostic tell was horizontal error exceeding vertical with a healthy χ^2 — geometrically impossible for this constellation (Section 3.2), and therefore a bug in the measurement of the measurement.

11.5 A window sized by the wrong bound

Discussed in Section 6.1: sizing the correlation search by geometry alone ignores that the peak also carries the clock offset difference. This failure is *time-dependent* — a fresh run looks perfect and degrades as drift accumulates — so a short smoke test cannot detect it. Measured after 26 minutes of running: clock deltas reaching 92 μs against a 27 μs window, all 104 fixes overflowing, $\chi^2_{\mathrm{red}} \approx 1.1 \times 10^6$, horizontal error 870 m.

12 Measured performance

Ten emulated nodes, two bursts per second, CP-OFDM emitter, moving target, disciplined clocks:

Quantity	**Value**
Aggregate ingest	3.7–4.4 Mbps
Correlate and solve (CPU, 21 weighted CAF pairs)	$\approx$ 100 ms
Reduced chi-square, median	0.45
Horizontal error, median	0.2 m
Vertical error, median	$\approx$ 2 m
Covariance containment (horizontal / vertical)	62 % / 65 % (target 68 %)

Selected controlled comparisons, each with everything else held fixed:

Change	**Without**	**With**
Sinc vs. parabolic interpolation	0.081 samp. RMS	0.0017 samp. RMS
CAF on a moving emitter (per pair)	26.7 ns worst	1.4 ns worst
CAF, full solve	1.7 m, $\chi^2 = 5.0$	0.2 m, $\chi^2 = 0.37$
All-pairs GLS vs. star reference	4.33 m mean	3.54 m mean
Robust loss, one corrupted node	330.6 m	8.9 m
GCC weighting, node-local interference	8.9 ns RMS	3.5 ns RMS
GCC weighting, fleet (defences off)	31.1 m	2.3 m
GCC weighting, fleet (defences on, p90)	3.19 m	0.56 m
Cyclostationary vs. energy detection	+8 dB	−2 dB
Channelizer vs. hopping emitter (detection)	1 of 16	16 of 16
Bandwidth under multipath (2 $\to$ 18 MHz)	4.57 m	0.26 m
Robust loss under multipath (no outlier)	4.57 m	4.50 m

Two of these deserve comment, in opposite directions.

Weighting's apparent value depends entirely on what else is running: in isolation it rescues a fix from 31 m to 2.3 m, but with the quality gate, robust loss and all-pairs GLS already absorbing single-pair damage, its contribution shrinks to a tail effect. Reporting the isolated figure as the system benefit would be dishonest.

The last row is a null result and is listed deliberately. Robust loss is the strongest defence in this document against a corrupted node (330.6 $\to$ 8.9 m) and is worth nothing at all against multipath, because one failure is an outlier and the other is common-mode. A table of improvements that omitted the case where a technique does nothing would misrepresent when to reach for it.

Note on comparability. The headline figures above are measured against a simulator that models the carrier, emitter motion, disciplined clock noise and a structured OFDM waveform. Earlier revisions of this system reported 0.4 m horizontal against a baseband-only simulator with a spectrally flat emitter. The current numbers are therefore not merely better; they are obtained under conditions the earlier configuration would not have survived — with the carrier modelled and the cross-ambiguity search disabled, the same fleet degrades to 1.7 m and $\chi^2_{\mathrm{red}} = 5.0$. Interference, hopping and multipath remain disabled by default, and each is documented above with the cost of enabling it.

A caution about where these numbers are taken. Solve time is a property of the machine as much as of the algorithm, and the failure mode is worth stating because it is indistinguishable from a regression. On a four-core mobile processor the full simulation — twelve processes, plus a browser drawing a city — can drive the package to its thermal limit, at which point the cores clock down to a fraction of nominal. Observed: package at 96 °C, cores pinned at 200–400 MHz against a 4700 MHz maximum, and a 64 kB FFT taking 40 ms on an *otherwise idle* machine where it takes 1.2 ms cool — a factor of 34. Solve time went from 86 ms to several seconds, the correlator fell permanently behind the burst rate, and the tracker began correcting with seconds-old fixes, which on screen looks exactly like a solver that has lost its target.

The tell is that the degradation is *uniform*: correlation, IQ transfer and an unrelated numerical benchmark all slow by the same factor, where a genuine regression is localised to whatever changed. The remedy is to let the machine cool and re-measure, never to retune the estimator against thermally-limited numbers. Cool, the same reduced configuration sustains 86–93 ms with every burst solved.

13 First hardware: a TinyB210 on the bench

Every number in §12 comes from emulated nodes. This section records what happened when a single real receiver — a HamGeek TinyB210, a B210 clone built on the AD9361 — was attached, and separates what that measurement established from what it could not.

13.1 What the device reports

Identity	`B210`, serial `271512`, EEPROM name `MyB210`
Firmware / FPGA	`8.0` / `16.0`, board revision 4, USB 3
Discipline	`GPSTCXO v3.9 for TinyB210`
Clock sources	`internal`, `external`, `gpsdo`
Time sources	`none`, `internal`, `external`, `gpsdo`
Sensors	`gps_gpgga`, `gps_gprmc`, `gps_time`, `gps_locked`, `gps_servo`, `ref_locked`
Receive	2 channels, 50–6000 MHz, up to 56 MHz bandwidth

That `gpsdo` appears as both a clock and a time source, and that `gps_locked` and `ref_locked` exist as sensors, is the wiring TDOA requires; a clone that had economised here would have been unusable. One detail matters later: the discipline is a *TCXO*, not an OCXO.

13.2 What was verified without a GPS antenna

Property	Result	Bearing on the theory
Requested vs. actual rate	exact at 2.4 and 20 Msps	a mismatch is a multiplicative bias on
Requested vs. actual f_c	exact at 2437 MHz	as above
Continuous 20 Msps, 2 ch	156 MB/s, no overflow	the wideband profile of §4.3 is affordab
Burst capture, isolated	clean, 8/8 and 12/12	the capture primitive itself is sound
Under full node workload	clean at 2.4 Msps; intermittent overflow at 20	host-side contention, not a link ceiling
Channel coherence, no antenna	0.030	independent noise, as §8 predicts
Channel coherence, 2 antennas	0.59 mean, 0.92 peak	the interferometer of §8 is real

Two results deserve emphasis. First, with two antennas the coherence between channels rose by more than an order of magnitude, which is the physical precondition for the two-element bearing estimate — the phase difference is meaningless if the channels are not coherent. Second, the cyclostationary detector fired on genuine over-the-air Wi-Fi, classifying it `uas_ofdm` and retaining bursts down to -7.8 dB SNR where the energy gate requires $+8$ dB. Wi-Fi is CP-OFDM, structurally what the detector was designed against, so this is the first confirmation on a signal no simulation produced that the $\sim$10 dB margin claimed in §5.2 is real.

13.3 What changes when the GPSDO locks

Without lock the device time source stays `internal`: device time counts from power-on with no relation to UTC, `time_aligned` is false, and `node.py` refuses `capture.mode: scheduled` outright rather than producing buffers that look simultaneous and are not. Four things change on lock.

1. The device clock is stepped to UTC on a PPS edge. `set_time_next_pps` — not `set_time_now` — writes the next whole UTC second so that the device's notion of time advances on the *physical* pulse. All nodes then agree to within their PPS accuracy rather than to within host scheduling jitter, which is hundreds of microseconds and would dominate everything in §6.

2. Scheduled capture becomes legal, and needs no coordination. Each node derives its capture instants from UTC alone, $t_k = k/f_{\text{burst}}$, so independently started nodes open the same window without exchanging a message; the slot index doubles as the burst id the supernode groups on. This replaces the bench fallback in which every node free-runs and inter-node TDOA is meaningless.

3. Frequency discipline bounds the correlation window. This is the quantitative one. Two oscillators with fractional frequency error $\Delta f/f$ accumulate relative timing offset

$$\Delta t(T) \;=\; \frac{\Delta f}{f}\,T, \tag{42}$$

and the correlation peak sits at geometric TDOA *plus* that offset, of which only the first term is bounded by the array baseline. With a $\pm$5 km baseline the geometric term is under $\pm 17\,\mu$s and the search window is $\sim 27\,\mu$s; (42) says how long each discipline mode keeps the true peak inside it:

Mode	Typical $\Delta f/f$	Time to exceed $27\,\mu$s
`free` (undisciplined TCXO)	10^{-6}	$\sim$27 s
`holdover` (TCXO, post-lock)	10^{-7}	$\sim$4.5 min
`gpsdo` (locked)	10^{-11}	$\sim$31 days

These are order-of-magnitude figures from oscillator specifications, not measurements of this board. The structure is what matters: GPS lock is what makes a *narrow* correlation window sustainable, and a narrow window is what buys the sidelobe rejection that §6.1 shows cannot simply be traded away by widening the search.

This is the same failure recorded in §6.1, seen from the clock side. It is also why `holdover` on *this* board deserves its own measurement: the discipline is a TCXO, and a TCXO's holdover is roughly an order of magnitude worse than the OCXO the mode name suggests. Losing GPS on this hardware is a matter of minutes before the window overflows, not hours.

4. The schedule error becomes observable. With a common UTC grid, the per-capture quantity $\mathrm{clk\_off} = -(t_{0,\mathrm{actual}} - t_{\mathrm{slot}})$ is the hardware's own report of how far its capture landed from the instant requested, and `sched_err_ns` — currently null — is populated. Its *standard deviation* across captures is the jitter figure below.

13.4 What remains unmeasured

The decisive number is untouched. A *constant* schedule offset is common to all nodes and cancels in the difference (2); the *jitter* does not, and enters the solution as timing noise amplified by geometry. At the dilution figures of §3.2 a jitter of σ_t contributes roughly $\mathrm{VDOP} \cdot c\,\sigma_t$ to vertical error, so 50 ns of jitter is already ∼240 m of vertical uncertainty before anything else is counted. That is why the acceptance threshold is where it is, and it cannot be evaluated without an antenna that can see the sky.

14 Limits

Emission is required. Everything here localizes emitters. Non-cooperative is not the same as non-emitting: consumer drones stream video continuously, which is why this approach is viable at all. A fully radio-silent autonomous drone is invisible to TDOA, to direction finding, and to the cross-ambiguity function alike. That regime needs passive radar against broadcast illuminators, acoustics, or electro-optical sensing.

That fallback is a different band, not merely a different mode. A 10-inch carbon-fibre propeller resonates as a half-wave dipole near 600 MHz, measured drone radar cross-section peaks there and falls some 6 dB by 1 GHz, and every published passive detection of a small drone uses a UHF illuminator — DVB-T, DAB or LTE450, roughly 450–770 MHz — never the 2.4 GHz band this system operates in. Reported ranges are 0.5–3 km against small airframes, the outlier being 5 km with a 50 kW transmitter; median airframe RCS is −20 to −15 dBsm with a further 5–9 dB of fluctuation loss. Any such extension therefore needs its own antennas, its own receive chain, and a direct-path canceller — the direct signal runs roughly 40 dB above the target return — none of which is shared with the emitter-localization path described here.

Node survey error propagates one-to-one. A single GNSS fix is accurate to 3–5 m, and node position error enters the solution directly. Real deployment requires RTK or a 24-hour static survey.

Clock error is systematic, not noise. The chi-square scaling in (33) honestly inflates covariance for random error but still under-reports bias-driven error. An uncalibrated deployment produces a confident, stable, wrong solution — the most dangerous failure mode available.

Detection thresholds are synthetic. The cyclostationary threshold and the classification boundary are set against simulated data. Their mechanics are proven; the constants require real band captures.

Single target. See data association above.

Hardware timing is still unmeasured. §13 closed part of this gap and not the decisive part. A real TinyB210 now streams at both profile rates without loss, tunes and samples at exactly the requested parameters, and produces coherent dual-channel data and genuine cyclostationary detections. What it has not done is lock its GPSDO, because that needs an antenna with sky view — so the schedule jitter remains unquantified. A constant offset is common to all nodes and cancels in (2); jitter does not. Below 50 ns the approach is viable; above 200 ns the timing path is unusable for TDOA and the hardware is useful only for direction finding. No amount of the mathematics in this document changes that number.

One receiver is not a fix. A single node contributes zero TDOA rows: the measurement is a *difference* between nodes, and min_nodes = 5 is a property of the geometry, not a configurable convenience. Bench work with one radio can validate the signal chain, the detector, the bearing estimate and the transport, and can validate nothing about localization.